

Digital Evidence Management Plan – Parrot OS

Goal: Collect live data, perform basic memory/disk forensics, maintain chain of custody, and create an event timeline.

1. Live Data Collection (System State, Processes, Network)

 Take screenshots of each command output.

System State

```
uname -a          # Kernel version, OS info
uptime            # System uptime
who -a            # Logged-in users
```

Running Processes

```
ps aux > ~/Evidence/running_processes.txt
less ~/Evidence/running_processes.txt # Open to show output
```

Network Connections

```
ss -tulpn > ~/Evidence/network_connections.txt
less ~/Evidence/network_connections.txt
```

You now have a snapshot of **system state + processes + network activity** saved to `~/Evidence/`.

2. Basic Memory Capture & Analysis

Memory Dump (Acquisition)

```
sudo dd if=/dev/mem of=~/.Evidence/memdump.raw bs=1M
ls -lh ~/.Evidence/memdump.raw # Verify size of file
```

Install & Run Volatility

Volatility is usually pre-installed in Parrot OS, but if not:

```
sudo apt update && sudo apt install volatility
```

Run a basic analysis (change profile if needed):

```
volatility -f ~/.Evidence/memdump.raw
--profile=LinuxUbuntu_5_10_0_amd64 pslist >
~/.Evidence/volatility_pslist.txt
```



Screenshot the volatility output (or open the saved file and screenshot a snippet).

3. Disk Acquisition (Forensic Imaging)

Use `dd` to make a forensic image of your VM's main disk.

```
sudo fdisk -l # Identify disk (e.g., /dev/sda)
sudo dd if=/dev/sda of=~/.Evidence/disk_image.dd bs=4M status=progress
```

Generate hash to prove integrity:

```
sha256sum ~/.Evidence/disk_image.dd > ~/.Evidence/disk_image_hash.txt
```



Screenshot:

- `fdisk -l` output (showing disk)
 - `sha256sum` result (hash value)
-

4. Chain of Custody Documentation

Create a simple text file documenting:

- **Date/time** evidence was collected
- **Who** collected it (your name)
- **Where** it is stored (path)
- **Hash values** to prove integrity
- **Purpose** (e.g., incident response investigation)

Example:

Chain of Custody Record

Collector: Matthew Rich

Date/Time: 2025-09-18 13:00

Evidence: memdump.raw, disk_image.dd, running_processes.txt

Hashes:

memdump.raw: [insert hash]

disk_image.dd: [insert hash]

Storage Location: ~/Evidence/

Purpose: Containment and forensic investigation of suspected compromise



Screenshot or include as an attachment in your report.

5. Timeline Creation (macOS + Parrot OS Logs)

You need to **correlate events** from both systems (host & guest VM).

Parrot OS Logs

```
journalctl --since "2025-09-18 00:00" --until "2025-09-18 23:59" >
~/Evidence/parrot_logs.txt
```

macOS Logs (on your host Mac)

Run in Mac Terminal:

```
log show --start '2025-09-18 00:00:00' --end '2025-09-18 23:59:59' > ~/Desktop/macOS_logs.txt
```

Then manually combine key events into a timeline table in your report, for example:

Time (UTC)	System	Event Description
10:15:03	Parrot	Interface enp0s1 taken down (isolation)
10:16:12	macOS	VirtualBox state saved (snapshot)
10:20:45	Parrot	Captured memory dump

```
yaml

Chain of Custody Record
-----
Collector: Matthew Rich
Date/Time: 2025-09-18 13:00
Evidence: memdump.raw, disk_image.dd, running_processes.txt
Hashes:
  memdump.raw: [insert hash]
  disk_image.dd: [insert hash]
Storage Location: ~/Evidence/
Purpose: Containment and forensic investigation of suspected compromise
```

Time (UTC)	System	Event Description
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10:15:03 Parrot Interface enp0s1 taken down
(isolation)

10:16:12 macO VirtualBox state saved
S (snapshot)

10:20:45 Parrot Captured memory
dump

```
bash: /home/user/Evidence/volatility_pslist.txt: No such file or directory
[~]-[user@parrot]-[~]
→ $sudo fdisk -l
Disk /dev/vda: 64 GiB, 68719476736 bytes, 134217728 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: B12C8E5C-F5A0-D14E-AF1A-184C19EFEB9E
```